

EBN122 CLASS TEST / KLASTOETS 1 (12 AUG 2016)**ANSWER SHEET / ANTWOORD BLAD**

Surname:	MEMO.	Student number:							
Initials:		AFR (Eng 3-6)	ENG (Eng 3-1)	ENG (Eng 3-6)					

Total:	15	/15
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1	a) $Q = 3.888 \text{ kC}$ ✓		b) $W = 13,608 \text{ kJ}$ ✓	
2	$V_1 = -6 \text{ V}$ ✓	$V_2 = 16 \text{ V}$ ✓	$V_3 = -24 \text{ V}$ ✓	
3	a) $R_{ab} = 25 \Omega$ ✓✓		b) $R_{cd} = 20 \Omega$ ✓✓	
4	$a = 2$ ✓	$b = 4$ ✓	$c = -1$ ✓	$d = 1$ ✓
5	$I_S = -7.5 \text{ A}$ ✓		$V_0 = 5 \text{ V}$ ✓	

DO NOT remove this page from the test.

EBN122 Class Test / Klastoets 1 (12 August 2016)**15 Marks/Punte****30 Minutes/Minute****Question 1 [2]**

A rechargeable flashlight battery is capable of delivering 60 mA for about 18h.

- How much charge can it provide over that period? (Answer in kC.)
- If the terminal voltage is 3.5V, how much energy can the battery deliver? (Answer in kJ.)

Vraag 1 [2]

'n Herlaaiibare flitsligbattery kan 60 mA vir 18h lewer.

- Hoeveel lading voorsien dit oor daardie periode? (Antwoord in kC.)
- Indien die terminaalspanning 3.5V is, hoeveel energy kan die battery lewer? (Antwoord in kJ.)

$$a) \quad Q = 60 \text{ mA} \times 18 \text{ h} \times 3600 \text{ s/h} = 3,888 \text{ kC.}$$

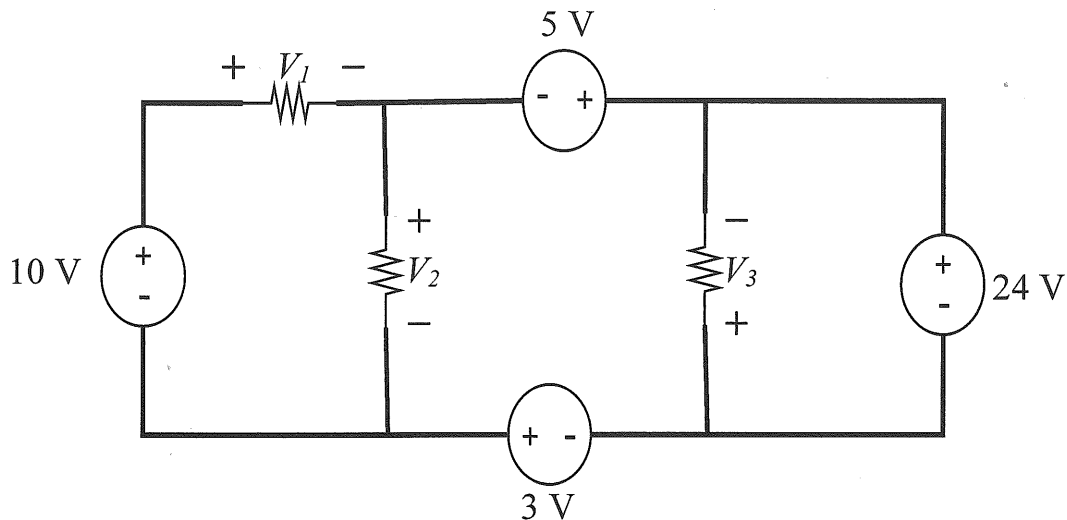
$$b) \quad W = P \cdot \Delta T = V \cdot I \cdot \Delta T = 3.5 \times 60 \text{ mA} \times 18 \text{ h} \times 3600 \text{ s/h} \\ = 13,608 \text{ kJ}$$

Question 2 [3]

For the circuit below, find V_1 , V_2 and V_3 .

Vraag 2 [3]

Vir onderstaande kringbaan, bepaal V_1 , V_2 en V_3 .



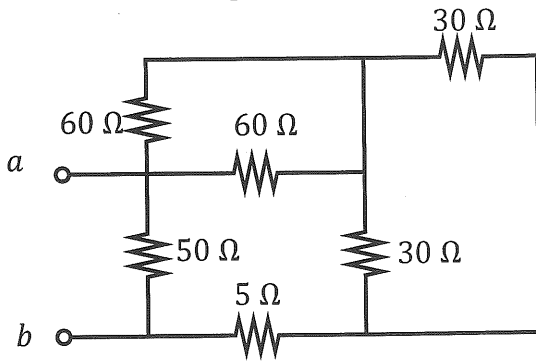
$$\text{KVL:} \quad +V_3 + 24 = 0 \Rightarrow V_3 = -24 \text{ V}$$

$$\text{KVL:} \quad -10 + V_1 - 5 + 24 - 3 = 0 \Rightarrow V_1 = -6 \text{ V}$$

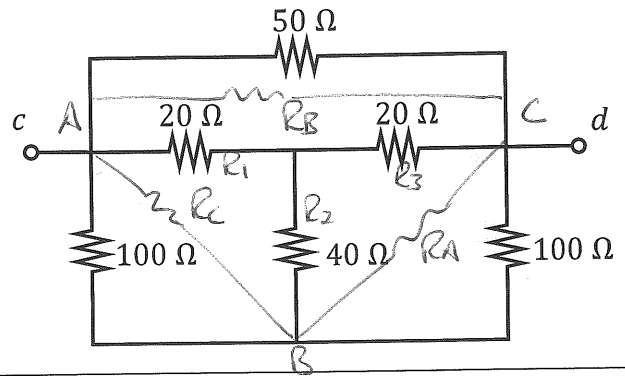
$$\text{KVL:} \quad -V_2 - 5 + 24 - 3 = 0 \Rightarrow V_2 = 16 \text{ V}$$

Question 3 [4]

- a) For circuit 1, determine R_{ab} .
b) For circuit 2, determine R_{cd} .

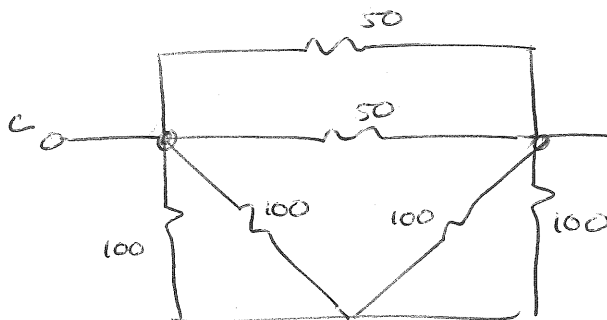
Kringbaan/Circuit 1

Vraag 3 [4]

- a) Vir kringbaan 1, bepaal R_{ab} .
b) Vir kringbaan 2, bepaal R_{cd} .

Kringbaan/Circuit 2


$$\begin{aligned}
 a) R_{ab} &= (30 \parallel 30 + 60 \parallel 60 + 5) \parallel 50 \\
 &= (15 + 30 + 5) \parallel 50 \\
 &= 50 \parallel 50 \\
 &= 25 \Omega \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 b) R_A &= \frac{20 \times 20 + 20 \times 40 + 40 \times 20}{20} = R_C \\
 &= \frac{2000}{20} = 100 \Omega. \\
 R_B &= \frac{2000}{40} = 50 \Omega
 \end{aligned}$$



$$\begin{aligned}
 R_{cd} &= (100 \parallel 100 + 100 \parallel 100) \parallel (50 \parallel 50) \\
 &= (50 + 50) \parallel 25 \\
 &= 100 \parallel 25 \\
 &= \frac{100 \times 25}{125} = 20 \Omega \quad \checkmark
 \end{aligned}$$

Question 4 [4]

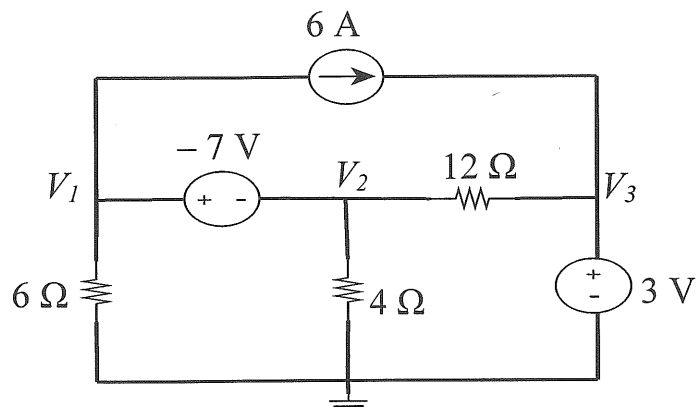
Apply nodal analysis to setup two equations of the following form:

$$\begin{aligned} aV_1 + bV_2 &= -69 \\ cV_1 + dV_2 &= 7 \end{aligned}$$

Vraag 4 [4]

Gebruik node analyse om twee vergelykings van die volgende vorm op te stel:

$$\begin{aligned} aV_1 + bV_2 &= -69 \\ cV_1 + dV_2 &= 7 \end{aligned}$$



$$* V_1 - V_2 = -7$$

$$* -V_1 + V_2 = 7$$

$$* \frac{V_1}{6} + \frac{V_2}{4} + \frac{V_2 - 3}{12} + 6 = 0$$

$$2V_1 + 3V_2 + V_2 - 3 = -72$$

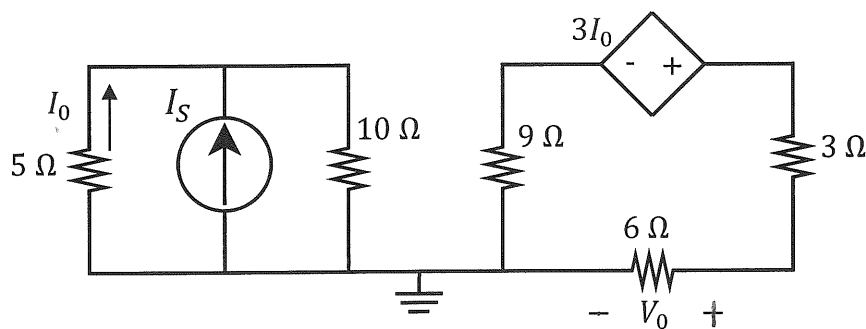
$$2V_1 + 4V_2 = -69$$

Question 5 [2]

If $I_0 = 5$ A, determine I_S and V_0 .

Vraag 5 [2]

Indien $I_0 = 5$ A, bepaal I_S en V_0 .



$$* I_0 = -\frac{I_S \times 10}{15}$$

$$I_S = \frac{-5 \times 15}{10}$$

$$= -7.5 \text{ A}$$

$$V_0 = \frac{+ 3(5) \times 6}{9 + 6 + 3}$$

$$= \frac{90}{18} = 5 \text{ V}$$